Microfluidic Device for Vascular Tissue Engineering Scaffolds

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BIPORES for Vascular Tissue Engineering

- **Problem:** Traditional vascular scaffolds fail to include the interconnected architecture necessary for cells to communicate and develop into mature tissues.
- **Bicontinuous Interfacially Jammed Emulsion (BIJEL):** Highly interconnected emulsion with a nanoparticle-stabilized interface.
- **BIJEL-Integrated Porous Engineered Systems (BIPORES):** Synthetic BIJEL-based biomaterial scaffolds with an interconnected porous structure and nanotopology.
- **Solution:** Provide a perfusable, vessel-like scaffold architecture to support transport, cellular communication, and vascular tissue maturation.

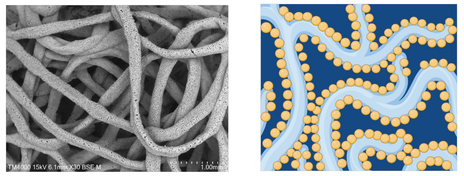


Figure 1: Process for BIJEL fiber formation. SEM image of BIPORES fibrous scaffold and phase-separated BIJEL network, illustrating the PEGDA phase, water phase, and the jammed nanoparticle interface.

Device Objective

Engineering Motivation: Current microfluidic devices cannot produce continuous hollow BIPORES fibers, creating a critical gap in translating this novel biomaterial for artificial vascular applications.

Objective	Target	Purpose
Fiber Geometry	100-300 µm bore diameter[2] 200 - 500 µm outer diameter[3]	Replicates relevant dimensions of vascular structures.
Continuous Extrusion	15 minutes	Continuous fibrous scaffolds for advanced applications.
Ease of Assembly	25 minutes	Fast, reproducible, and user-friendly assembly.

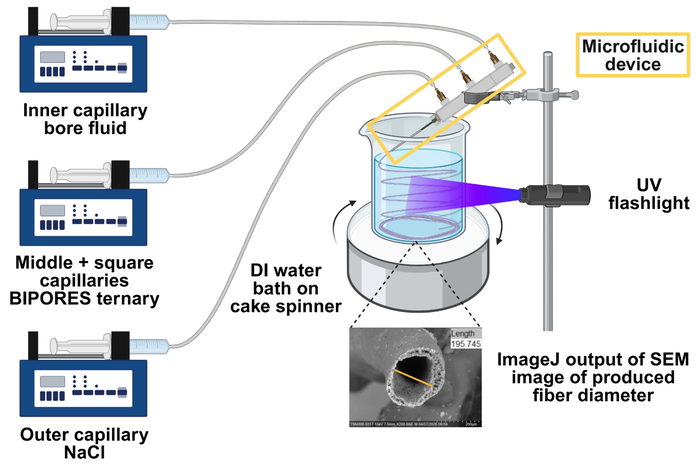


Figure 2: Device setup for hollow BIPORES fiber extrusion.

Device Design and Fabrication

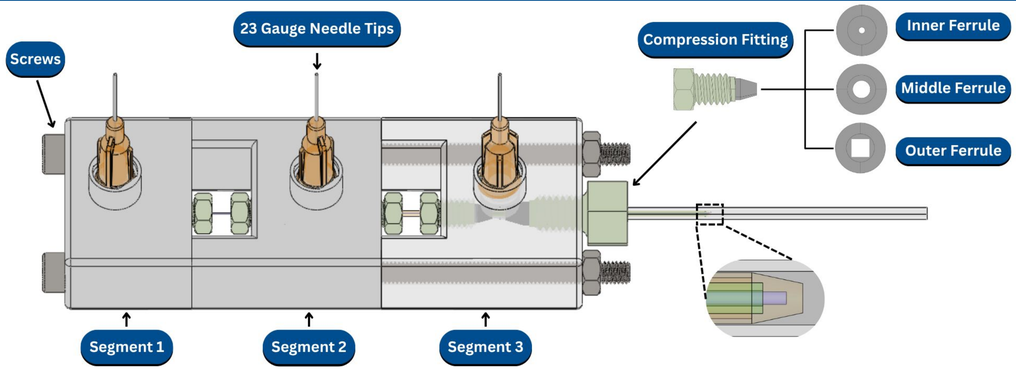


Figure 3: CAD models of the assembled device and components. Compression fittings provide leak tight sealing and hold capillaries in place. Screws ensure a stable device connection, and 23-gauge blunt needle tips for fluid inlet. Insert shows the alignment of four glass capillaries.

Rail System Produced 3x Faster Assembly Time

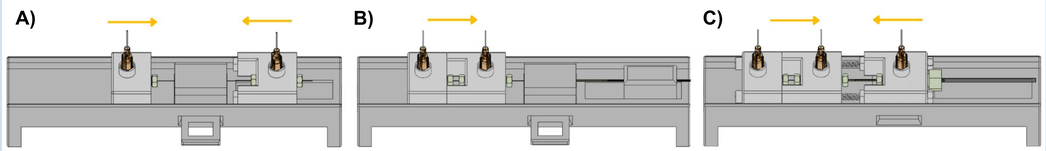


Figure 4: Assembly process workflow. Rail-guided device three-step assembly. (A) Inner capillary is positioned inside Segments 1 and 2 together using inner capillary alignment jig. (B) Middle capillary positioned and secured in Segment 2 using middle capillary alignment jig. (C) Segment 3 attached to complete device using rails. Rail device reduces assembly time by 3x compared to manual assembly.

Successful Hollow Fiber Formation and Geometry Validation

Achieved >80% Fiber Geometry Within Target Ranges

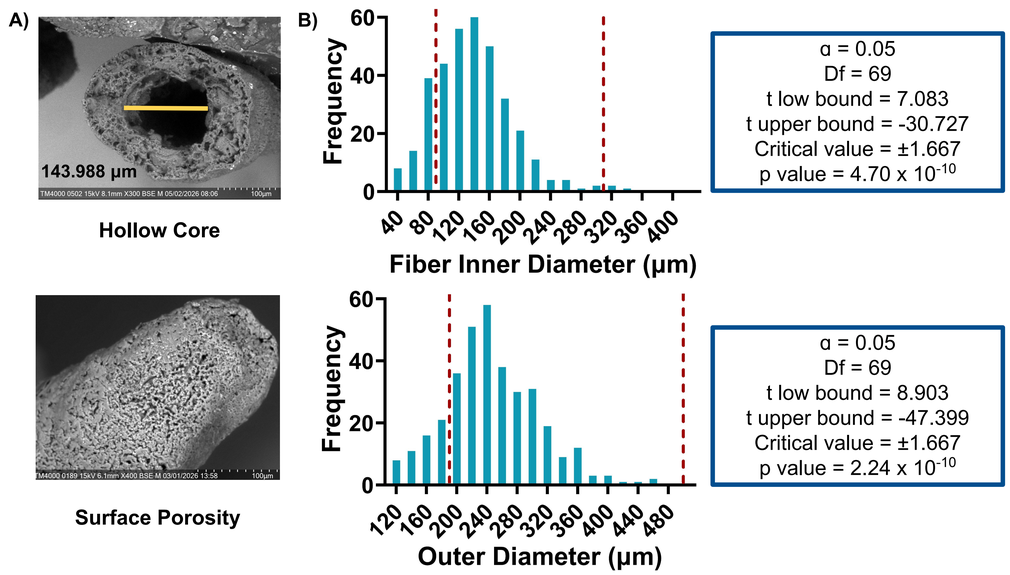


Figure 5: Macrostructural observation of BIPORES fibers and microstructure analysis via Scanning Electron Microscopy (SEM). (A) Cross section (top), side view (bottom). (B) Fiber geometry analysis via one sample equivalence test.

Key Outcomes

- ✔ Successfully designed **reusable, modular** microfluidic device capable of producing hollow-core BIPORES fibers.
- ✔ Demonstrated **continuous extrusion** for up to 20 minutes.
- ✔ Validated **100%** hollow-core fiber formation through SEM.
- ✔ Fabricated hollow fibers within target dimensional ranges:
  - ✔ Inner diameter: 100-300 µm
  - ✔ Outer diameter: 200-500 µm
- ✔ Confirmed **successful formation** of porous fiber architectures.

Next Developments

- Integrate a UV curing module at the device tip to enable controlled polymerization.
- Achieve 30-second mounting time onto a custom bioprinter for rapid device setup and switching.
- Seed scaffold with endothelial cells to promote lumen formation.

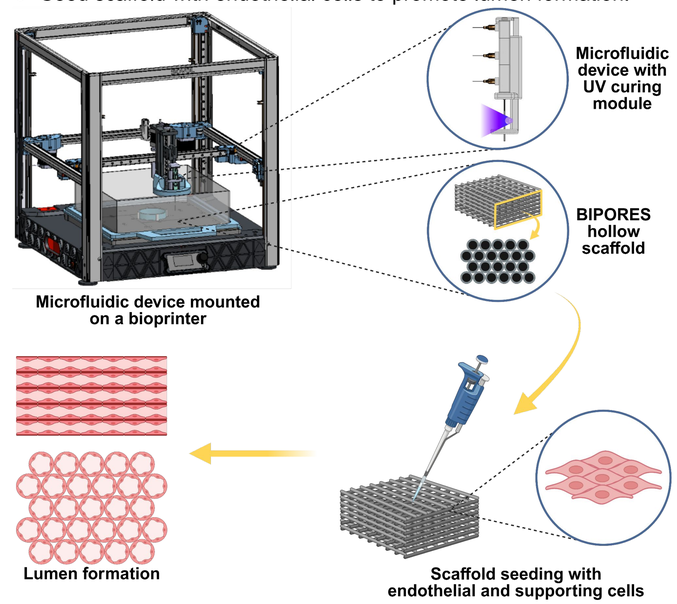


Figure 6. Microfluidic device with integrated UV curing module mounted onto a custom made bioprinter for continuous fiber extrusion. Extruded scaffold is seeded with endothelial and supporting cells to fabricate synthetic vascular structures.

Acknowledgements and References

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